Finance Manager Dashboard

Team Implementation Assignment

Project:	PhysioGuidance eLearning Platform
Client:	Roya Almustaqbal for Health Training
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▪▪ BLOCKER: Tap is the only gateway available for active testing. Moyasar, Tamara, and Tabby are Saudi/GCC region-locked — Amna must create sandbox accounts using Saudi credentials before these can be tested. Development must proceed using MockGateway for all non-Tap gateways until credentials are received.

5 Widgets	4 Gateways	10 Sprints	NELC Compliant

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1. Assignment Overview & Objective

The Finance Manager Dashboard is a critical M4 deliverable for the PhysioGuidance platform. It provides the Finance Manager with complete visibility and control over all financial operations — revenue tracking, payment processing, invoice management, refund workflows, discount analytics, and profitability reporting. The platform must support four Saudi/GCC payment gateways and comply fully with NELC, PDP Law, and OWASP standards.

Core Objectives:

- Build a gateway-agnostic payment backend that supports Tap, Moyasar, Tamara, and Tabby using a unified ProviderPattern — so adding or swapping gateways requires minimal code change.
- Implement MockGateway for all local development and testing of non-Tap gateways until Saudi sandbox credentials are available.
- Deliver all 5 Finance Manager Dashboard widgets with full Arabic/English bilingual support (RTL/LTR), date-range filtering, drill-down, and CSV/PDF export.
- Ensure all financial event logs are immutable and exportable for NELC audit evidence.
- Enforce PDP Law data protection: no PII in payment error logs, data retention policies applied.

2. Payment Gateway Status & Strategy

All 4 Gateways | Current Access & Build Approach

All four payment gateways are Saudi/GCC region-locked. Tap is the only gateway with active sandbox access. The team must use a MockGateway for Moyasar, Tamara, and Tabby until Amna provides sandbox credentials. The gateway-agnostic architecture ensures zero wasted development effort.

Gateway	Type	Status	Dev Priority	Sandbox Access	Notes
Tap Payments	Card / Mada / Apple Pay	**ACTIVE – Testing**	**P1 – Build Now**	Available (shared keys)	Primary gateway; implement full flow first
Moyasar	Card / Mada / STC Pay	**PENDING**	**P2 – Parallel Build**	Requires Amna credentials	Build with MockGateway; swap when access granted
Tamara	BNPL (Buy Now Pay Later)	**PENDING**	**P2 – Parallel Build**	Requires Amna credentials	10% deduction rule on refunds must be enforced
Tabby	BNPL (Buy Now Pay Later)	**PENDING**	**P3 – After Tamara**	Requires Amna credentials	Similar flow to Tamara; reuse BNPL provider pattern

MockGateway Strategy:

- MockGateway is a local implementation of the ProviderPattern interface that simulates all real gateway responses.
- It supports all payment states: success, failed, pending, refunded — configurable per test scenario.
- All frontend components and business logic can be fully tested using MockGateway before real gateway credentials arrive.

- Switching from MockGateway to a real gateway is a single configuration change — no business logic changes required.
- MockGateway also serves as the baseline for automated unit and integration tests that run in CI/CD.

3. Dashboard Widgets — What to Build

5 Core Widgets | Finance Manager Dashboard

The Finance Manager Dashboard consists of five primary widgets. Each widget must support Arabic/English toggle, date range filters (Today / 7D / 30D / Custom), drill-down, and CSV/PDF export. All data must be real-time or near-real-time.

● Widget 1 — Revenue Summary

- Display total revenue: daily, weekly, monthly — with trend vs. previous period.
- Average Order Value (AOV) calculated and displayed per time range.
- Gateway comparison: side-by-side revenue breakdown for Tap, Moyasar, Tamara, and Tabby.
- BNPL vs direct payment split — show % of revenue from each payment type.
- Chart: line chart for revenue trend; bar chart for gateway comparison.
- Filter by: date range, course, category, payment method, language.

● Widget 2 — Payments & Invoices

- Table of all transactions: columns — Invoice #, Learner, Course, Amount, Gateway, Status, Date.
- Status filter: Paid / Pending / Failed — with error code display for failed payments.
- For failed payments: show gateway error code (e.g., insufficient_funds, card_expired, 3ds_failed).
- Actions: Resend Invoice (email), Download Invoice PDF, View Transaction Detail.
- Invoice PDF: branded, bilingual, includes course name, learner details, gateway, amount, tax.
- Real-time status updates — pending payments refresh automatically.
- Pagination, search by learner name / invoice number, export full list as CSV or PDF.

● Widget 3 — Refunds

- Refund queue: shows all refund requests with status (Pending Approval / Approved / Rejected / Processed).
- Each refund entry shows: requester, course, original amount, refund reason, request date.
- Finance Manager can Approve or Reject with a mandatory reason/note field.
- Tamara BNPL refund rule: automatically deduct 10% processing fee before applying refund.
- Refund reason categories: Policy Request, Course Cancelled, Duplicate Payment, Technical Error, Other.
- Approved refunds trigger gateway API refund call automatically.
- Full audit trail: every approve/reject action logged with timestamp and Finance Manager user ID.

● Widget 4 — Discounts & Coupons

- Create coupon: code, discount type (flat SAR / percentage %), max uses, expiry date, applicable courses.
- Coupon list: shows code, usage count vs. max, total discount value given, active/expired status.
- Usage analytics: lift measurement — revenue with coupon vs. without; cannibalization detection.
- Coupon performance chart: conversions attributed to each coupon code.
- Disable/activate coupon toggle with immediate effect.
- Bulk coupon generation for cohorts (e.g., 50 unique codes for a specific course batch).

● Widget 5 — Financial Reports

- Profitability report by Course: revenue, refunds, net profit, margin % per course.
- Profitability report by Instructor: revenue generated, instructor fees, net contribution.
- Profitability report by Delivery Type: On-site vs. Live (Zoom) vs. Blended.
- Invoice aging report: outstanding invoices grouped by age (0-30d, 31-60d, 61-90d, 90d+).
- Chargeback rate report: by course and payment gateway.
- Export formats: Excel (.xlsx) with formatted sheets, and PDF with Gamma Developers branding.
- All exports include: timestamp, filters used, generated by (user ID), NELC evidence metadata.

4. Architecture Overview

Gateway-Agnostic Provider Pattern

The payment backend must use a provider/strategy pattern so the system is not tightly coupled to any single gateway. This is the same pattern established during earlier PhysioGuidance backend planning. Each gateway implements the same interface; the rest of the application never calls a gateway directly.

Provider Pattern — Key Components:

- **PaymentProvider Interface**: Defines the contract — charge(), refund(), getStatus(), createWebhook(). Every gateway class implements this interface.
- **TapProvider**: Full implementation using Tap Payments API. Active for testing now.
- **MoyasarProvider**: Implementation ready; activated when sandbox credentials received from Amna.
- **TamaraProvider**: BNPL implementation with 10% refund deduction logic built in.
- **TabbyProvider**: BNPL implementation; reuses Tamara pattern with Tabby-specific API calls.
- **MockGateway**: Fake implementation for development — returns configurable success/failure responses.
- **PaymentService**: Orchestrator that selects the correct provider based on the payment method chosen at checkout.
- **WebhookHandler**: Receives and verifies gateway callbacks; updates payment/invoice status in DB.
- **InvoiceEngine**: Generates invoice records and PDF files on payment confirmation.

■ Architecture Principle: The frontend Finance Dashboard never knows which gateway processed a payment. It only calls internal PhysioGuidance API endpoints. This protects against vendor lock-in and makes future gateway additions (e.g., STCPay standalone) a backend-only change.

5. Core Database Tables

Finance Module — Key Entities

The following tables are the minimum required for the Finance Manager Dashboard. The team must implement these migrations as the first task in Sprint 1.

payments

Column	Type	Description
id	UUID PK	Unique payment identifier
user_id	UUID FK	References users table
course_id	UUID FK	References courses table
invoice_id	UUID FK	References invoices table
gateway	ENUM	tap | moyasar | tamara | tabby | mock
gateway_tx_id	VARCHAR	Transaction ID from the gateway
amount	DECIMAL(10,2)	Total charged amount in SAR
status	ENUM	pending | paid | failed | refunded | chargeback
error_code	VARCHAR	Gateway error code if failed (e.g., insufficient_funds)
payment_method	ENUM	card | mada | apple_pay | bnpl
created_at	TIMESTAMP	When payment was initiated
updated_at	TIMESTAMP	Last status change

invoices

Column	Type	Description
id	UUID PK	Unique invoice identifier
invoice_number	VARCHAR	Human-readable invoice number (INV-2025-00001)
user_id	UUID FK	Learner who was invoiced
course_id	UUID FK	Course purchased
subtotal	DECIMAL(10,2)	Amount before discount
discount_amount	DECIMAL(10,2)	Coupon/discount applied
total	DECIMAL(10,2)	Final amount invoiced
status	ENUM	draft | issued | paid | overdue | cancelled
pdf_url	VARCHAR	S3/storage URL to the generated invoice PDF
issued_at	TIMESTAMP	Invoice issue date
due_at	TIMESTAMP	Payment due date

refunds

Column	Type	Description
id	UUID PK	Unique refund record
payment_id	UUID FK	References payments table
requested_by	UUID FK	User who requested the refund
approved_by	UUID FK	Finance Manager who approved
amount_requested	DECIMAL(10,2)	Full refund amount requested

Column	Type	Description
amount_refunded	DECIMAL(10,2)	Actual amount after deductions (e.g., Tamara 10%)
deduction_amount	DECIMAL(10,2)	Gateway fee deducted
deduction_reason	VARCHAR	e.g., Tamara BNPL 10% processing fee
reason	ENUM	policy | cancelled | duplicate | technical | other
status	ENUM	pending | approved | rejected | processed
notes	TEXT	Approver notes (mandatory on reject)
created_at	TIMESTAMP	Request date

coupons

Column	Type	Description
id	UUID PK	Unique coupon record
code	VARCHAR UNIQUE	Coupon code (e.g., HEALTH2025)
discount_type	ENUM	flat | percentage
discount_value	DECIMAL(10,2)	SAR amount or % value
max_uses	INTEGER	Maximum number of uses (NULL = unlimited)
used_count	INTEGER	Increments on each successful application
applicable_courses	UUID[]	Array of course IDs; empty = all courses
expires_at	TIMESTAMP	Expiry date/time; NULL = no expiry
is_active	BOOLEAN	Finance Manager can toggle on/off
created_by	UUID FK	Finance Manager who created it
created_at	TIMESTAMP	Creation date

7. Sprint Plan

10-Week Delivery Schedule — Finance Module

The sprint plan is structured to unblock Tap development immediately while building all other gateways behind MockGateway. Real gateway activation happens in Sprint 4 once credentials are received.

Sprint	Duration	Focus Area	Key Deliverables	Priority
Sprint 1	Week 1–2	Foundation + Tap Integration	MockGateway, ProviderPattern, Tap full flow (charge/webhook/refund), invoice generation, payments DB schema	Critical
Sprint 2	Week 3–4	Dashboard UI + Revenue Module	Revenue Summary widget, Payments & Invoices table, real-time status updates, gateway comparison charts	Critical
Sprint 3	Week 5–6	Refunds + Coupons Module	Refund queue with approval workflow, Tamara 10% rule, coupon CRUD, usage analytics, lift tracking	High
Sprint 4	Week 7–8	Moyasar + BNPL Integration	Moyasar provider, Tamara & Tabby BNPL flows, sandbox swap (when credentials received), full regression	High
Sprint 5	Week 9–10	Financial Reports + QA	Profitability reports by course/instructor/delivery, Excel/PDF export, NELC audit export, full QA pass	Medium

■ Sprint 4 (Moyasar + BNPL) is timeline-dependent on Amna providing sandbox credentials. If credentials arrive earlier, Sprint 4 can begin immediately. The MockGateway approach ensures Sprint 4 is already 70% complete by the time credentials arrive — only the gateway class and configuration swap remain.

8. Acceptance Criteria

Definition of Done — Finance Dashboard

All items below must pass before the Finance Dashboard is considered complete and ready for client review. The QA Engineer is responsible for verifying each criterion.

#	Acceptance Criterion	Verification Method
1	Revenue Summary shows correct daily/weekly/monthly totals per gateway	Compare DB totals with dashboard display; test with seeded data
2	Tap full payment flow: charge → webhook → invoice generated → PDF downloadable	End-to-end test with Tap sandbox; verify invoice PDF content
3	Failed payments display correct error codes (e.g., insufficient_funds, card_expired)	Trigger each error scenario in Tap sandbox; verify UI error code display
4	Refund queue: Finance Manager can approve/reject; Tamara applies 10% deduction	Create refund request; approve as Finance Manager; verify deduction math

#	Acceptance Criterion	Verification Method
5	Coupon created, assigned to course, applied at checkout, usage count increments	Full coupon lifecycle test; verify lift % calculation in analytics
6	Financial Report: profitability by course/instructor/delivery exported as Excel and PDF	Generate report; verify data accuracy; download both formats
7	MockGateway simulates all gateway states (success, failure, pending)	Unit tests covering all MockGateway states
8	Dashboard bilingual: all labels, dates, currency display correctly in AR and EN	Switch language; verify RTL layout; verify SAR currency format
9	All financial data exports include timestamp, filters used, and user context (NELC)	Export CSV/PDF; verify header metadata
10	No PII exposed in payment error logs; audit log immutable	Review error log format; attempt log modification

9. NELC Compliance Requirements

Finance Module — Mandatory Standards

The Finance Dashboard must satisfy the following NELC and regulatory compliance requirements. Non-compliance blocks platform certification.

- **Audit Trail (Immutable Logs)**: Every financial event (payment, refund, coupon use, invoice generation) must be logged in an append-only audit table. No record may be modified or deleted. Logs must be exportable as NELC evidence.
- **PDP Law Data Protection**: Payment error logs must not contain PII (no card numbers, no full names in error fields). Learner financial data must be accessible via Subject Access Request. Data retention policies must be configurable.
- **Bilingual Financial Documents**: All invoices, financial reports, and exported documents must be available in Arabic and English. Arabic invoices must use RTL layout with SAR currency notation.
- **WCAG 2.1 AA — Dashboard UI**: Finance Dashboard must meet color contrast requirements, support keyboard navigation, and all chart data must have accessible text alternatives. Screen reader compatibility required.
- **OWASP Security — Payment Flows**: All payment API endpoints must require authentication. Webhook endpoints must verify gateway signatures. No payment secrets in frontend code. Rate limiting on payment attempts.
- **CSV/PDF Evidence Exports**: Every widget export must include: generation timestamp, filters applied, generating user ID, and platform version. This metadata maps to NELC evidence bundle requirements.

10. Key Dependencies & Blockers

Actions Required to Unblock Development

BLOCKED — Amna Action Required	Moyasar, Tamara, Tabby sandbox accounts must be created by Amna using Saudi business credentials. Share API keys with dev team to activate Sprint 4. MockGateway covers all development until received.
Dev Team — Sprint 1 Start	Implement DB migrations (payments, invoices, refunds, coupons tables), MockGateway class, ProviderPattern interface, and TapProvider. This unblocks all other parallel tracks.
Frontend — Design Tokens	Finance Dashboard must use the established PhysioGuidance design system CSS variables. No new color variables or font sizes unless approved. RTL layout must be tested before each PR merge.
QA — Sandbox Test Plan	QA must prepare test scenarios for all Tap sandbox error codes before Sprint 2 ends. MockGateway test cases must cover all states (success, failed, pending, refund) with automated tests.
DevOps — Environment Variables	Tap API keys must be in environment variables only — never hardcoded or committed. Set up separate .env files for dev (MockGateway), staging (Tap sandbox), and production.